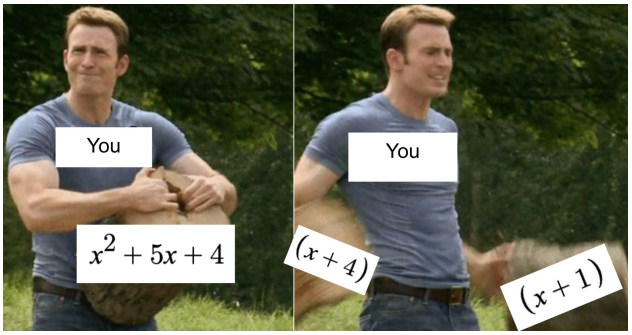

Grade 12 Advanced Functions
Factor Theorem Test

First Name: _____

Last Name: _____

Directions:

- Please answer to 2 decimal points
- The test is designed to be completed in 75 minutes

For grading use only

Page:	2	3	4	5	Total
Points:	10	8	8	4	30
Score:					

Multiple Choice (6 marks)

1. (1 point) Which value of a makes $(x - a)$ a factor of $f(x) = x^3 + 2x^2 - x - 2$?
   A. -2
   B. -1
   C. 0
   D. 1
2. (1 point) If $f(x) = x^3 - 7x + 6$, which of the following is NOT a factor?
   A. $(x - 1)$
   B. $(x - 2)$
   C. $(x + 3)$
   D. $(x + 2)$
3. (1 point) What is the remainder when $f(x) = 2x^3 - 5x^2 + 7x - 1$ is divided by $(x - 2)$?
   A. -1
   B. 0
   C. 7
   D. 9
4. (1 point) Which binomial is a factor of $f(x) = x^3 + 3x^2 + 2x$?
   A. $(x - 2)$
   B. $(x - 1)$
   C. $(x + 2)$
   D. $(x + 3)$
5. (1 point) If $f(x) = x^3 - x^2 - 4x + 4$, which of the following is a root?
   A. -2
   B. 0
   C. 2
   D. 1
6. (1 point) Which value of k makes $(x + 3)$ a factor of $f(x) = 2x^3 + kx^2 + 5x - 3$?
   A. -3
   B. 0
   C. 6
   D. 8

Long Answer (24 marks)

7. **Factor Theorem Application**
   (a) (4 points) Is $(x-2)$ a factor of $f(x) = x^3 - 3x^2 + 4$? Justify your answer using the Factor Theorem.

8. **Finding a Coefficient for a Factor**
   (a) (4 points) Find a value of k such that $(x+3)$ is a factor of $f(x) = 2x^3 + kx^2 + 5x - 3$.

9. **Factoring a Cubic Polynomial**
   (a) (4 points) Factor $f(x) = x^3 - 7x + 6$ completely.

10. **Determining a Root**

    (a) (4 points) Determine if $x = -2$ is a root of $f(x) = x^3 + 3x^2 + 2x$.

11. **Remainder Theorem**

    (a) (4 points) What is the remainder when $f(x) = 2x^3 - 5x^2 + 7x - 1$ is divided by $(x - 2)$?

12. **Factoring with Integer Roots**

(a) (4 points) Factor $f(x) = x^3 + x^2 - 4x - 4$ completely.